Shreya S Rao

AI Engineer | LLMs, RAG & Python Backend

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Summary

AI Engineer building production LLM and RAG systems for hospitality. Built a hotel website chat assistant that handles **20,000+ guest conversations a month**, and re-architected its retrieval to hybrid search, lifting answer accuracy from **75% to 95%** and cutting latency by **40%**. Strong Python backend foundation (Django, DRF, PostgreSQL) with Docker and AWS deployment.

Education

Dayananda Sagar College of Engineering *Bengaluru, India*
B.E. in Computer Science and Engineering | GPA: 9.2/10 *2021 – 2025*

Experience

Zapcom Solution Pvt. Ltd. *Bengaluru, India*
AI Engineer I *Jul 2025 – Present*

- Built an **LLM-powered website chat assistant** for a hospitality SaaS platform, serving single hotels and chains (18+ properties for one group) and handling **20,000+ guest conversations per month**, cutting manual support tickets by **∼30%**.
- Re-architected the RAG layer from FAISS + OpenAI embeddings to **hybrid retrieval on ChromaDB** (vector + BM25 with RRF fusion, LLM query expansion, cross-encoder reranking, semantic cache), raising answer accuracy from **75% to 95%** and cutting latency by **40%**.
- Built incremental ingestion with domain-aware semantic chunking, a custom chunker for section-formatted knowledge bases, and SHA-256 manifest diffing so only changed content is re-embedded (local BGE embeddings).
- Developing **multi-agent orchestration** with strict knowledge-base grounding, hospitality-domain fallback and hotel comparison, upsell and cross-sell flows.
- Built backend services and OpenAI-driven automations (ticket updates, NL-to-structured-data queries) with **Django, DRF and PostgreSQL**, and owned **AWS EC2 + Docker** deployment, monitoring, logging and incident handling.

Zapcom Solution Pvt. Ltd. *Bengaluru, India*
Associate Software Engineer (Internship) *Jan 2025 – Jul 2025*

- Built and enhanced REST APIs for internal and client-facing applications using Python, Django, Flask and PostgreSQL.
- Supported data ingestion and processing workflows using Python and SQL, strengthening API design and database fundamentals.

NIRAMAYA | IIT Bhilai *Bengaluru, India*
Python Developer Intern *Sep 2023 – Mar 2024*

- Developed an AI-driven system to help underprivileged individuals access medical emergency loans.
- Designed ML-based algorithms to score patient eligibility and loan risk from health and financial indicators and recommend optimal loan amounts.
- Presented project outcomes at IIT Bhilai and contributed to research documentation under academic mentorship.

Projects

Real-Time Fire Detection & Alerting System / *YOLO, OpenCV, Python • Hackathon*

- Built a real-time fire detection pipeline using a YOLO-based CNN with frame-wise inference on live camera streams, rendering bounding boxes with confidence scores.
- Added an event-driven email alerting subsystem to notify stakeholders on detection for rapid incident response.

Scrum Master Automation / *Python, Jira REST API • Hackathon*

- Automated sprint operations: generated Jira user stories from CSV, split them into tasks, assigned sprints and managed status transitions, reducing manual planning effort.

Mirror Mate / *Raspberry Pi, OpenCV, Linux*

- Built an AI-powered smart mirror (weather, news, calendar, web dashboard) with OpenCV-based emotion detection that recommends personalised music.

Technical Skills

Languages: Python, SQL
GenAI: OpenAI GPT-4, RAG, Hybrid Search (Vector + BM25, RRF), Cross-Encoder Reranking, Embeddings (OpenAI, BGE), Semantic Chunking, Semantic Caching, Multi-Agent Orchestration, Prompt Engineering
ML & Vector DBs: ChromaDB, FAISS, PyTorch, TensorFlow, Keras, scikit-learn, OpenCV
Backend: Django, Django REST Framework, Flask, REST APIs, Microservices, Celery, Redis
Cloud & Data: AWS (EC2), Docker, PostgreSQL, MySQL, MongoDB, Data Pipelines
Tools: Git, Postman, Streamlit, Jira, Agile/Scrum
Certifications: Complete SQL Bootcamp, 100 Days of Code (Python Pro), Data Visualization Using Python, Cloud Foundations, MongoDB